



Garlock 706

MATERIAL PROPERTIES*

Color:	White
Composition:	Inorganic fibers with a nitrile binder
Fluid Services¹:	Saturated and superheated steam ³
Temperature², °F (°C)	
Minimum:	-100 (-73)
Continuous Max:	+750 (+399)
Maximum:	+1000 (+538)
Pressure², Maximum, psig (bar):	1500 (104)
P x T (max.)², psig x °F (bar x °C)	
1/32 and 1/16":	700,000 (25,000)
1/8":	500,000 (18,500)
Meets Specification:	ABS (American Bureau of Shipping) and Fire Safe

PHYSICAL PROPERTIES*

ASTM F36	Compressibility, range, %:	7-17
ASTM F36	Recovery, %:	50
ASTM F38	Creep Relaxation, %:	18
ASTM F152	Tensile, Across Grain, psi (N/mm²):	1400 (9)
ASTM F1315	Density, lbs./ft.³ (grams/cm³):	105 (1.68)
ASTM F433	Thermal Conductivity (K), W/m²K (Btu.in./hr.ft.².°F):	0.29-0.38 (2.00-2.65)
ASTM D149	Dielectric Properties, range, volts/mil.	
	Sample conditioning	<u>1/16"</u> <u>1/8"</u>
	3 hours at 250°F:	133 142
	96 hours at 100% Relative Humidity:	25 25
ASTM F586	Design Factors	<u>1/16"</u> <u>1/8"</u>
	"m" factor:	11.4 ⁽⁴⁾ 22 ⁽⁴⁾
	"y" factor, psi (N/mm ²):	4800 (33.1) 6500 (44.8)
ROTT	Gasket Constants, 1/16":	Gb=2,455 a=0.267 Gs=0.622
ASTM F104	Line Call Out:	F712102A9B3E34K5L501M9 ⁽⁵⁾

SEALING CHARACTERISTICS*

	ASTM F37B Fuel A	ASTM F37B Nitrogen
Gasket Load, psi (N/mm²):	500 (3.5)	3000 (20.7)
Internal Pressure, psig (bar):	9.8 (0.7)	30 (2)
Leakage	0.5 ml/hr.	4.0 ml/hr.

IMMERSION PROPERTIES* - ASTM F146 Fluid Resistance after Five Hours

	ASTM #1 Oil 300°F (150°C)	ASTM IRM #903 300°F (150°C)	ASTM Fuel A 70-85°F (20-30°C)	ASTM Fuel B 70-85°F (20-30°C)
Thickness Increase, (%)	0-10	0-15	0-15	0-20
Weight Increase, (%)	<15	-	<20	<20
Tensile Loss, (%)	-	<55	-	-

Notes:

This is a general guide and should not be the sole means of selecting or rejecting this material. ASTM test results in accordance with ASTM F-104; properties based on 1/32" (0.8mm) sheet thickness unless otherwise mentioned.

* Values do not constitute specification Limits

¹ See Garlock chemical resistance guide.

² Based on ANSI RF flanges at our preferred torque. When approaching maximum pressure, continuous operating temperature, minimum temperature or 50% of maximum P x T, consult Garlock Applications Engineering. Minimum temperature rating is conservative.

³ Minimum recommended assembly stress = 4,800psi. Preferred assembly stress = 6,000-10,000psi. Gasket thickness of 1/16" strongly preferred. Retorque the bolts/studs prior to pressurizing the assembly. For saturated steam above 150psig, consult Garlock Engineering.

⁴ This "M" value, based on ambient temperature leakage with nitrogen, is high. Field experience has shown that lower values would be workable in elevated temperatures. Consult Applications Engineering.

⁵ A9: Leakage in Fuel A (Isooctane), Gasket Load = 500psi (3.5N/mm²), Pressure = 9.8psig (0.7bar): Typical = 0.5ml/hr, Max = 1.5ml/hr. M9: Tensile Strength = 1,400psi min. (9.7N/mm² min.).